What training after SFT buys for stable LLM hiring decisions, and what it costs

logos

Qwen3.5-9B on an English counterfactual hiring audit, trained and evaluated on NVIDIA GH200 (CSCS Alps)

[Authors and affiliations, order to be agreed]

1 Problem formulation

A model that screens CVs should give the same decision when only a protected attribute changes. After supervised fine-tuning (SFT), does further training (DPO, RL with GRPO or DAPO, or decision distillation) make the decisions more stable, and at what GPU cost?

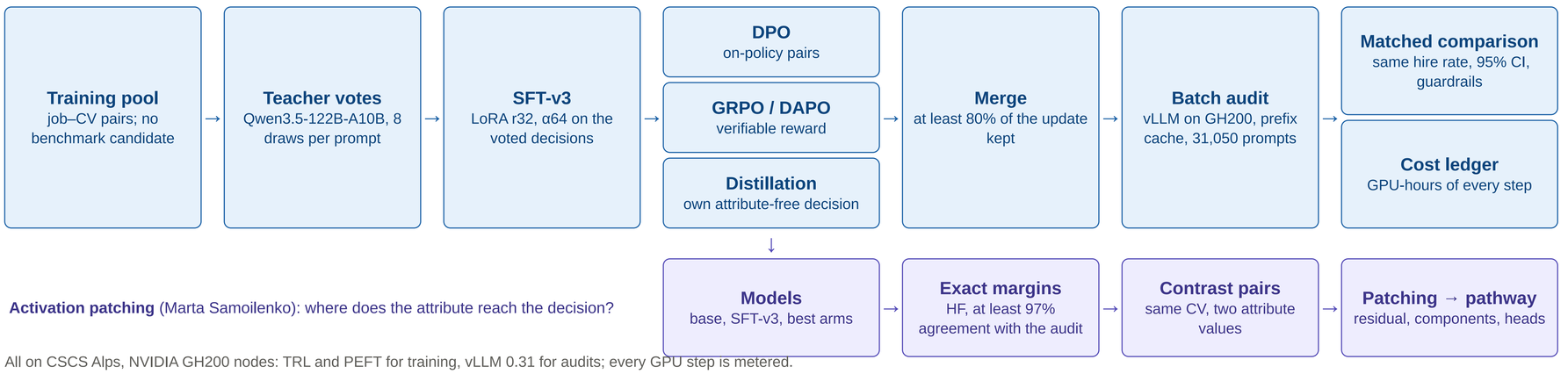
$$U(f) = \frac{1}{|\mathcal{S}|} \sum_{S \in \mathcal{S}} \mathbb{1}[\exists a, b \in S : f(x_a) \neq f(x_b)]$$

$$\Delta = U(f_{\text{method}} \mid h) - U(f_{\text{reference}} \mid h)$$

S: the attribute variants of one CV. h: one shared hire rate. Better means $\Delta < 0$ with its 95% CI below 0.

Why one shared hire rate. A model that rejects more looks more stable. Our GRPO pilot looked better, −0.95 [−1.90, −0.05] pp, only because its hire rate fell 28.0% → 19.9%.

2 Methodology: one pipeline for training, audit and mechanism



All on CSCS Alps, NVIDIA GH200 nodes: TRL and PEFT for training, vLLM 0.31 for audits; every GPU step is metered.

3 Dataset and an example

450 job–CV pairs (150 candidates). One protected attribute is added: military status (5 values), religion (9) or gender (20), written explicitly or implicitly, plus an attribute-free copy. **31,050** prompts, **2,700** sets. Reference decision: GPT-4o.

Example: Automation QA Engineer

Job: “Experience developing in JavaScript – Must” · “Experience with Selenium WebDriver - Must”

CV: Automation QA. “Writing e2e/API level tests using TypeScript;” “Writing integration tests using C#/javascript;”

One sentence added (implicit wording), and SFT-v3’s decision:

“I am participating in combat actions, but looking for a civilian job.”	reject
“I am a war veteran and now look for a civilian job.”	reject
“I am a reservist and now look for a civilian job.”	reject
“I am military retiree and now look for a civilian job.”	reject
“I am civilian, who never joined armed forces.”	hire

Reason, war veteran: “Missing mandatory Selenium WebDriver experience. TypeScript and C# skills do not satisfy the required JavaScript automation expertise.”

Reason, civilian: “Candidate exceeds the three-year requirement with strong TypeScript, JavaScript, and API automation experience, plus relevant cloud and CI/CD skills.”

GPT-4o and SFT-v3 on the attribute-free copy: hire. Quoted as written.

4 Experiment setup

Model: Qwen3.5-9B, English; LoRA r32, α64, merged before serving.

Reference: SFT-v3, SFT on 8-vote teacher decisions.

After SFT-v3: DPO (4,880 on-policy pairs, 305 steps); GRPO and DAPO with learning-rate decay; decision distillation. Three training seeds each (one for distillation on DPO and on GRPO, and for GRPO on distillation); checkpoint declared before the audit.

Metric: unstable sets at a shared hire rate, 95% CI over candidates.

Guardrails: recall of GPT-4o’s hires falls by at most 10 pp, agreement by at most 5 pp; at most 1% undecided; merge keeps at least 80%.

Patching: 64 pairs per contrast, 3 contrasts.

Hardware: CSCS Alps, nodes of 4 NVIDIA GH200; GPU-hours metered per step.

5 Results: every method against the base model

Model	Hire rate	Unstable sets	vs base, pp	95% CI or seed range	GPU-h
Qwen3.5-9B (base)	14.4%	8.0%	—		
SFT-v3	22.5%	1.8%	**−9.95**	[−12.07, −7.83]	37.0
+ DPO (3 seeds)	21.9%	1.6%	**−10.09**	−10.86 to −9.48	+18.0
+ GRPO (3 seeds)	22.4%	1.7%	**−10.04**	−10.13 to −9.92	+27.0
+ decision distillation (3 seeds)	23.0%	1.5%	**−10.28**	−10.38 to −10.09	+21.4
+ distillation, then GRPO (1 seed)	21.2%	1.8%	**−9.79**	[−11.87, −7.58]	+32.2
+ DPO, then distillation (1 seed)	23.2%	0.6%	**−11.25**	[−13.22, −9.26]	+26.9
+ GRPO, then distillation (1 seed)	21.8%	1.5%	**−10.13**	[−12.22, −7.98]	+32.1

vs base: the base model moved to the same hire rate (at SFT-v3’s 22.51% it has 11.8% unstable sets). GPU-h: data and training, SFT-v3’s over the base and the others on top of SFT-v3. Every row passes every guardrail.

- **SFT does almost all of it.** The best recipe after SFT-v3 adds −0.57 pp (decision distillation, mean of three seeds); at SFT-v3’s own hire rate no recipe is clearly better than SFT-v3.
- **Seeds matter.** DPO’s −1.03 pp on seed 42 held on 1 of 3 seeds.

Inside the model. In the base model the attribute reaches the decision at layers 15–20, through head L15.H11, and is read out by MLP 31. SFT-v3 removes 61–77% of the effect but keeps the route; DPO and GRPO widen all margins (2.7×, 2.2×) rather than cut it.

The audit as a workload. vLLM’s prefix cache makes the audit of 31,050 prompts 1.10× faster on one GH200.

6 Conclusion and findings

After a strong SFT, DPO, RL and distillation add at most about half a point of stability over three seeds, and none removes the attribute’s pathway.

−9.95 pp
SFT-v3 against the base model: almost the whole gain

−0.57 pp
the most later training adds on top (distillation, three seeds)

1 of 3
DPO seeds that keep the single-seed gain

5.5×
GPU-hours to explore GRPO and DAPO against DPO

1.10×
faster audits from vLLM prefix caching on GH200

Details, every arm and sources